

Problem Sheet

Geometric and Topological Data Domains

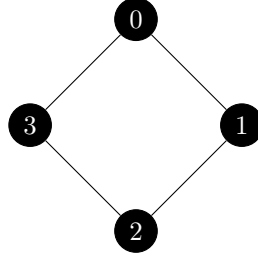
Topology and Geometry in Machine Learning

Lecture 2

Remark. The exercises are marked in increasing difficulty by *, **, ***.

Exercise 1 (Graphs, the Laplacian, and the Graph Fourier transform)

Let $G = (V, E)$ be the 4-cycle with $V = \{0, 1, 2, 3\}$ and $E = \{\{0, 1\}, \{1, 2\}, \{2, 3\}, \{3, 0\}\}$.



- (a, *) Write down the adjacency matrix A , the degree matrix D , and the graph Laplacian $L = D - A$ of G .
- (b, *) Choose an orientation of the four edges and write the signed incidence matrix $B \in \mathbb{R}^{4 \times 4}$. Verify that $L = BB^T$ and that $B^T \mathbf{1} = \mathbf{0}$ (recall that $\mathbf{1} = (1, 1, 1, 1)^T$).
- (c, **) Compute the eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \lambda_4$ of the Laplacian. Find the Fiedler value λ_2 and interpret it. Compute a basis of orthonormal (orthogonal and with unit norm) eigenvectors of the Laplacian $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$.
- (d, ***) For a general graph $G = (V, E)$ and a signal $\mathbf{f} \in \mathbb{R}^n$, prove the identity

$$\mathbf{f}^T L \mathbf{f} = \sum_{\{i,j\} \in E} (f_i - f_j)^2.$$

Deduce that L is positive semi-definite and that $\mathbf{f}^T L \mathbf{f} = 0$ if and only if $\mathbf{f}$ is constant on each connected component.

- (e, *) Let $\mathbf{f} = (4, 0, 0, 0)^T$ be a spike signal at vertex 0. Compute the Graph Fourier coefficients $a_i = \mathbf{u}_i^T \mathbf{f}$ and check that $\mathbf{f} = \sum_{i=1}^4 a_i \mathbf{u}_i$.
- (f, **) Apply the low-pass filter that keeps only the modes with $\lambda_i \leq 2$. Write the filtered signal $\mathbf{f}_{\text{lp}}$ in vertex coordinates and compare with $\mathbf{f}$. What do you get if you keep only the mode $\lambda_1 = 0$?

Exercise 2 (Simplicial complexes and homology)

For each simplicial complex X below, orient every simplex by increasing vertex label.

Linear algebra toolbox. The following linear algebra facts can be used throughout the exercise:

- for a quotient vector space V/W , $\dim V/W = \dim V - \dim W$;
- the rank of matrix is the number of linearly independent rows (or columns), in other words, the dimension of its image;
- $\text{rank } \partial_k + \dim \ker \partial_k = \dim C_k = |X_k|$ (from rank-nullity).

(a, *) **Hollow vs. filled triangle.** Let T° have vertices $\{0, 1, 2\}$ and edges $\{(0, 1), (1, 2), (0, 2)\}$ (no 2-simplex), and let T be T° together with the 2-simplex $(0, 1, 2)$.

(i) Write the boundary matrices ∂_1 (for both) and ∂_2 (for T) and verify $\partial_1 \partial_2 = 0$.

(ii) Compute H_0, H_1, H_2 and the Betti numbers $\beta_0, \beta_1, \beta_2$ of T° and of T .

(b, *) **Wedge of two circles.** Let W have

$$V = \{0, 1, 2, 3, 4\}, \quad E = \{(0, 1), (1, 2), (0, 2), (2, 3), (3, 4), (2, 4)\}$$

(two triangle boundaries glued at the vertex 2, nothing filled). Compute all Betti numbers of W .

(c, **) **Hollow tetrahedron.** Let $S = \partial\Delta^3$ be the boundary of the 3-simplex on $\{0, 1, 2, 3\}$: all 4 vertices, all 6 edges, all 4 triangles, no 3-simplex. Compute $\dim C_k(S)$, the ranks of ∂_1, ∂_2 , and all Betti numbers. Identify a generator of $H_2(S)$ explicitly as a chain.

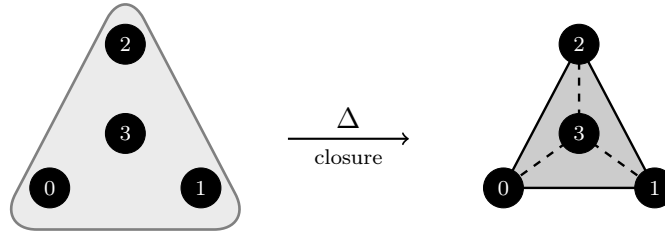
Exercise 3 (Simplicial complexes, hypergraphs, and closures)

Recall that a hypergraph is a pair $H = (V, \mathcal{E})$ with elements in $\mathcal{E}$ being non-empty subsets of V ; and that a (finite abstract) simplicial complex is a hypergraph that is *closed under inclusion*: $\sigma \in \mathcal{E}$ and $\emptyset \neq \tau \subseteq \sigma$ imply $\tau \in \mathcal{E}$. Define the *downward closure* (or *simplicial closure*) of H by

$$\Delta(H) := \{ \tau : \emptyset \neq \tau \subseteq h \text{ for some } h \in \mathcal{E} \},$$

or in other words, by adding all non-empty subsets of the hyperedges of H .

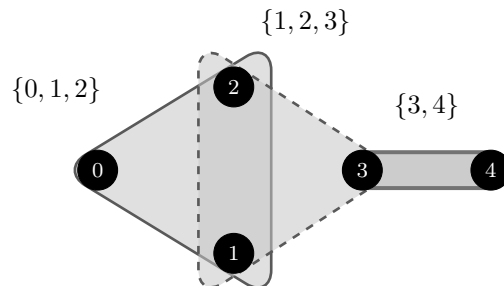
For example, for a hypergraph on 4 vertices with only one hyperedge $h = \{0, 1, 2, 3\}$, the closure is the full tetrahedron Δ^3 :



$$H = (\{0, 1, 2, 3\}, \{ \{0, 1, 2, 3\} \}) \quad \Delta(H) = \Delta^3 \text{ (all } 2^4 - 1 = 15 \text{ faces)}$$

(a, **) Show that $\Delta(H)$ is a simplicial complex, and that $\Delta(H) = H$ if and only if H is already a simplicial complex. Show also $\Delta(\Delta(H)) = \Delta(H)$.

(b *) Let $V = \{0, 1, 2, 3, 4\}$ and $\mathcal{E} = \{ \{0, 1, 2\}, \{1, 2, 3\}, \{3, 4\} \}$. List $\Delta(H)$ by dimension (X_0, X_1, X_2) and give its dimension.



(c, ***) **Universal property.** Prove that for every simplicial complex X on V ,

$$\Delta(H) \subseteq X \iff \mathcal{E} \subseteq X.$$

Conclude that $\Delta(H)$ is the smallest simplicial complex containing H .

(d, **) If a hyperedge h has $|h| = k$ vertices, how many simplices does it contribute to $\Delta(H)$? What would be more efficient, memory wise: to keep all simplices or to only keep the hyperedges?

Exercise 4 (Proving the Hodge decomposition theorem)

In the lecture, we have established that the boundary matrices of a complex satisfy $\partial_k \partial_{k+1} = 0$. This identity is the only thing one needs to prove the Hodge decomposition theorem.

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ be real matrices such that

$$AB = 0.$$

These can be seen as maps $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $B : \mathbb{R}^p \rightarrow \mathbb{R}^n$ between Euclidean spaces, where every space carries the standard inner product (so the adjoint of a matrix is its transpose), and we set

$$\Delta = A^\top A + BB^\top \in \mathbb{R}^{n \times n}.$$

Remark. If $A = \partial_k$ and $B = \partial_{k+1}$ are the boundary matrices of a simplicial complex, then $\Delta = L_k$ is the k -th Hodge Laplacian.

Linear algebra toolbox. You can use the following facts about matrices throughout the exercise. For $M \in \mathbb{R}^{r \times s}$ and $x \in \mathbb{R}^s$, $y \in \mathbb{R}^r$:

- $\langle Mx, y \rangle = (Mx)^\top y = x^\top M^\top y = \langle x, M^\top y \rangle$;
- *rank-nullity*: $\dim \ker M + \text{rank } M = s$ (number of columns), and $\text{rank } M = \text{rank } M^\top$;
- for any subspace $W \subseteq \mathbb{R}^s$, $\mathbb{R}^s = W \oplus W^\perp$ orthogonally;
- if $W \subseteq U$ are subspaces of $\mathbb{R}^s$, then $U = W \oplus (U \cap W^\perp)$ orthogonally.

Part 1: relating the image and kernel of $M \in \mathbb{R}^{r \times s}$ and its transpose $M^\top \in \mathbb{R}^{s \times r}$.

- (1, *) Prove $\ker M^\top = (\text{im } M)^\perp$ and $\ker M = (\text{im } M^\top)^\perp$.
- (2, *) Deduce the orthogonal splittings $\mathbb{R}^r = \text{im } M \oplus \ker M^\top$ and $\mathbb{R}^s = \text{im } M^\top \oplus \ker M$.
- (3, **) Prove $\ker(M^\top M) = \ker M$. (*Hint: if $M^\top Mx = 0$, look at $x^\top M^\top Mx$.*)
- (4, **) Using rank-nullity, prove $\text{rank}(M^\top M) = \text{rank } M^\top$ and hence $\text{im}(M^\top M) = \text{im } M^\top$.

Part 2: properties of two matrices with $AB = 0$.

- (5, **) Show $AB = 0 \iff \text{im } B \subseteq \ker A$. Taking transposes, show also $\text{im } A^\top \subseteq \ker B^\top$, and conclude $\text{im } B \perp \text{im } A^\top$.
- (6, **) For $x \in \mathbb{R}^n$ prove $x^\top \Delta x = \|Ax\|^2 + \|B^\top x\|^2$, and deduce

$$\ker \Delta = \ker A \cap \ker B^\top.$$

- (7, **) Prove the orthogonal splitting $\ker A = \text{im } B \oplus \ker \Delta$.

Hint: $\text{im } B \subseteq \ker A$ by (5); apply the last toolbox fact with $W = \text{im } B$, $U = \ker A$, then rewrite $(\text{im } B)^\perp$ and $\ker A \cap \ker B^\top$.

- (8, *) Prove likewise $\ker B^\top = \text{im } A^\top \oplus \ker \Delta$.

Hint: apply (7) verbatim to the pair $B^\top$, $A^\top$ in place of A , B ; note this does not change Δ .

(9, **) **Hodge decomposition.** Combine (2) (applied to B) with (8) to obtain the orthogonal direct sum

$$\mathbb{R}^n = \operatorname{im} A^\top \oplus \ker \Delta \oplus \operatorname{im} B$$

and, using that Δ is symmetric, $\operatorname{im} \Delta = \operatorname{im} A^\top \oplus \operatorname{im} B$.